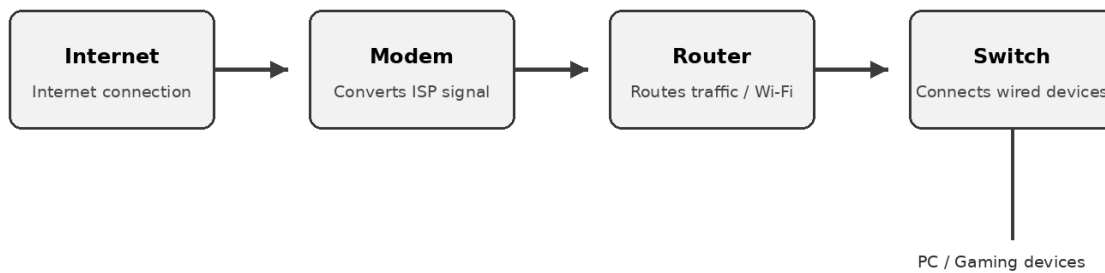


1) Router, Modem and Switch Setup

- Modem connects the home network to the ISP.
- Router routes traffic and provides Wi-Fi.
- Switch connects multiple wired devices on the local network.
- A typical connection is Internet → Modem → Router → Switch → Wired devices.



2) Connected Devices on a Home Wi-Fi Router

- Router administration page shows connected devices.
- Laptop — 192.168.1.10 — A4:5E:60:12:34:56.
- Phone — 192.168.1.11 — B8:27:EB:45:67:89.
- Gaming PC — 192.168.1.12 — 3C:52:82:98:76:54.

Home Wi-Fi Router — Connected Devices

Device	IP Address	MAC Address	Connection
Laptop	192.168.1.10	A4:5E:60:12:34:56	Wi-Fi
Phone	192.168.1.11	B8:27:EB:45:67:89	Wi-Fi
Gaming PC	192.168.1.12	3C:52:82:98:76:54	Ethernet

3) Router vs Modem vs Switch

- Router: routes traffic, is commonly used at home/office, can assign IP addresses using DHCP, and usually supports Wi-Fi.
- Modem: provides the connection between the local network and ISP; commonly used at home/office; normally does not assign LAN IP addresses; normally does not provide Wi-Fi.
- Switch: connects wired devices on a LAN; common in offices and gaming cafés; does not normally assign IP addresses; does not provide Wi-Fi.

Device	Main function	Used in	Assigns IP?	Wi-Fi?
Router	Routes traffic; connects network	Home / Office	Yes (DHCP)	Usually
Modem	Connects network to ISP	Home / Office	Usually no	No
Switch	Connects wired devices on LAN	Office / Gaming café	No	No

4) Gaming Café Network Setup

- 1 modem for the ISP connection.
- 1 router for Internet access, DHCP and Wi-Fi.
- 2 switches for wired LAN connections.
- 12 gaming PCs connected mainly through Ethernet for stable, low-latency connections.
- Wi-Fi is used for phones and staff devices.
- This setup provides both reliable wired gaming access and wireless connectivity.

